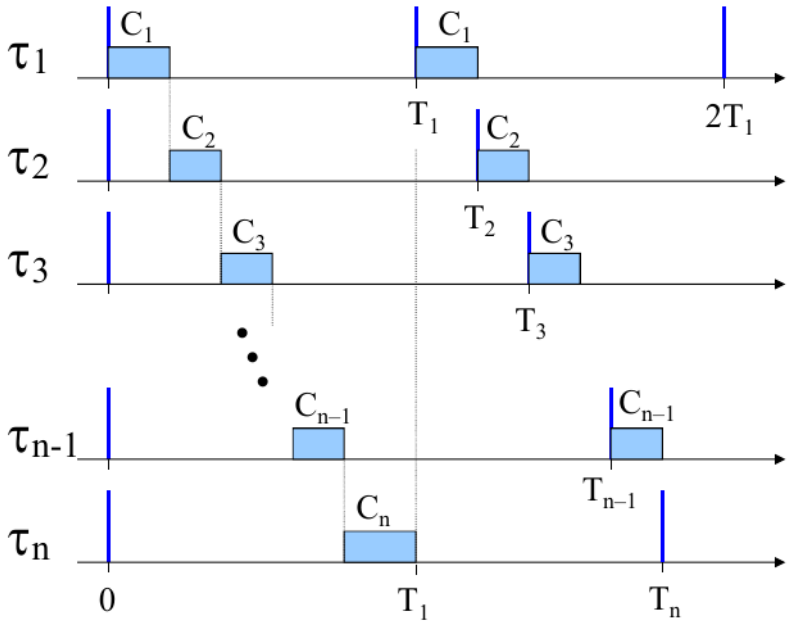
RM hyperbolic bound (Bini et al, 2003)

Theorem

If $\prod_{i=1}^{n}(U_i+1) \leq 2$ *for a set* Γ *of* n *pure periodic tasks* $\Rightarrow \Gamma$ *is schedulable by RM*

- Worst case conditions for the schedulabiliy of n tasks with $T_1 < T_2 < \cdots < T_n$:
  $T_n < 2T_1$, $C_1 = T_2 - T_1$, $C_2 = T_3 - T_2$, ..., $C_n = T_1 - (\sum_{i=1}^{n-1} C_i) = 2T_1 - T_n$

LO VISTO NELLA DIMOSTRAZIONE PER IL BOUND CON n TASK



DA 0 A T_1 VIENE ESEGUITO 1 JOB DI OGNI TASK USANDO AL 100% IL PROCESSORE.

Image by Prof. G. Buttazzo, Scuola Superiore Sant'Anna, Pisa

E. Bini, G. C. Buttazzo, G. M. Buttazzo, "Rate monotonic scheduling: The hyperbolic bound", IEEE Trans. on Comp., 52(7):933–942, July 2003

RM hyperbolic bound: proof

- $R_i := \frac{T_{i+1}}{T_i} \ \forall\, i \in \{1, \ldots, n-1\} \Rightarrow R_i = \frac{\overbrace{T_{i+1} - T_i}^{\text{C: PER LE CONZIONI DI CASO PESSIMO (VEDI FIGURA).}} + T_i}{T_i} = U_i + 1$ and $\prod_{i=1}^{n-1} R_i = \frac{T_n}{T_1}$

Proof.

- The worst case schedulability condition is $\overbrace{\sum_{i=1}^{n} C_i}^{\text{1 JOB PER OGNI TASK DA 0 A } T_1\text{. È IL CASO PESSIMO (VEDI FIG SOPRA).}} \leq T_1 \Rightarrow$

$$\Rightarrow \sum_{i=1}^{n-1} C_i + C_n \leq T_1 \Rightarrow C_n \leq T_1 - \sum_{i=1}^{n-1} C_i \Rightarrow C_n \leq 2T_1 - T_n \text{ given that}$$

$\overbrace{\sum_{i=1}^{n-1} C_i}^{\text{VEDI NELLA FIGURA}} = T_n - T_1$ by the worst case schedulability conditions $\Rightarrow$ DIVIDO TUTTO PER T_m

$$\Rightarrow \underbrace{U_n}_{\frac{C_m}{T_m}} \leq \frac{2T_1}{T_n} - 1 \Rightarrow U_n + 1 \leq \frac{2T_1}{T_n} = \frac{2}{\prod_{i=1}^{n-1} R_i} = \frac{2}{\prod_{i=1}^{n-1} (U_i + 1)} \Rightarrow$$

$$\Rightarrow \prod_{i=1}^{n} (U_i + 1) \leq 2$$

□